

Group 9 explainable AI (Task 10)

Introduction

To better understand how and why our model was predicting the classes it was predicting, we ran the outputs of the model through various explainable AI methods. This document describes the methods we used, the findings from each method, and the conclusions we pulled from the findings.

Part 1

Method

Part 1 of our explainable AI methods was the gradient X input method (gXi). The gXi method works by calculating the gradient of the derivation of the output and then multiplying it with the input.

Findings

The immediate thing we looked at was whether the same tokens produced different results. If this did occur, that meant the model was applying different values to tokens based on the context of the token, meaning the model was working correctly. We observed this occurring, and therefore the model was using context. An example of this can be seen in the image “*DISGUST Gradient X Input Explanation*”, where the token “No” is positive in one instance, and negative in the other.

The outputs of gXi also showed that our model was sometimes using the whole sentence, and sometimes just singular words in a sentence to classify it. In our error analysis, we already theorized the model was using singular words to classify sentences into wrong categories (the “Occlusion” category), and that seems to be confirmed here, with an example in “*ANGER Gradient X Input Explanation*” with the token “strawberry”.

A weird observation we made was present in a few of the graphs. As is seen in “*NEUTRAL Gradient X Input Explanation*” with the token “?”, it has both a positive and negative score. We do not know what is causing this phenomenon, and we do not see a pattern in which this is occurring, as it’s happening to both symbols and words.

Part 2

Method

In the second part, we used Layer-wise Relevance Propagation (LRP) to explain the model's outputs. In theory, this is supposed to spread what the model uses to come to an output, so every token is used instead of a very select amount.

Findings

We noticed that the model changed which tokens it used positively or negatively for each prediction, such as in the case of *"happiness - example 1 – predicted = anger"*, where all but 1 token have been changed to the opposite. This means that the changes brought to the model drastically changed what the tokens were being used for, instead of what we understood as only changing how many and how much each token was being used.

Another thing we noticed but are not exactly sure whether it is important is the fact that the LRP explanations were smaller than the gXi. LRP would often be below 0.1 while gXi could reach high numbers such as 1.25.

Part 3

Method

In the 3rd and final part, we would remove the least useful token (decided through gXi) each step and then see what the confidence of the model will be. An important value to keep in mind is 0,5 (50%). If the confidence of the model drops below that, it means the prediction is uncertain. We plotted the confidence per amount of tokens removed.

Findings

What we noticed early on is that the graphs did the opposite of what we expected to happen, such as in *"Model Robustness for: 'Here, here. What are you waiting for?'"*. With each removed token, the confidence went up. It is sometimes expected that when a sentence is fully used, the confidence of the model barely drops or sometimes goes up, such as in *"Model Robustness for: 'Jalapeno pepper, green, and red jalapeno pepper. Gift.'"* Such graphs mean the entire sentence is being used, which is very good.

We noticed that some of the confidences at 0 tokens removed were below at 50%, such as in *"Model Robustness for: 'You are for desserts, I don't look for anything for desserts, you look for everything for desserts.'"* This is bad, since that means the model is uncertain about some of the predictions it predicts, and means there is room for

improvement. Luckily, this seemed to happen only about 10% of the time, meaning the model was at least confident in the overwhelming majority of cases.

We also observed high fluctuations of the confidence of the model in some cases, such as in the case of “*Model Robustness for: “I won’t choose him either”*”, where the confidence goes from 80% to 43% to 100% to 39%.

Conclusion

In conclusion, the findings here seem to confirm some of our theories we’ve had, such as the model sometimes using very few tokens to achieve a prediction. This means we can better inform the client of what the limitations of our model are. Alongside that, our model seemed to sometimes go against what we had expected, meaning it changed our understanding of it a little bit.

Images







